PowerGrid Predictive Maintenance – Machine Learning Capstone Report

Predicting grid asset failure risk and prioritizing proactive maintenance

1. Executive Summary

This project develops a leakage-aware machine-learning framework for PowerGrid Utilities Corporation using the supplied PowerGrid_Utility_Intelligence.xlsx dataset. The objective is to estimate grid-failure risk, identify important failure drivers, and translate model output into an operational maintenance-priority framework.

The supplied workbook contains 50,500 rows and 32 columns. There are 500 exact duplicate records, leaving 50,000 unique records after deduplication. The binary target grid_failure_flag has a 43.02% failure rate. Missingness is concentrated in oil_quality_score (10.0%) and inspection_score (10.1%).

Four models were evaluated: Logistic Regression, Decision Tree, Random Forest, and SVM. Random Forest was selected because it achieved the strongest PR-AUC (0.886) and ROC-AUC (0.911) while maintaining 0.943 recall at its validation-selected operating threshold of 0.29.

2. Business Problem and Objectives

PowerGrid needs to move from reactive maintenance toward condition- and risk-based intervention. The analytical objectives are to predict grid_failure_flag, identify major failure drivers, rank assets by predictive risk, combine risk with customer exposure for maintenance prioritization, and provide interpretable recommendations.

3. Dataset Audit

rows: 50500.0

columns: 32.0

duplicate_rows: 500.0

failure_count: 21725.0

failure_rate_pct: 43.01980198019802

The workbook contains no reliable timestamp field, so a true historical-to-future temporal split could not be performed. The modeling strategy therefore uses stratified train/validation/test splits.

Data Dictionary

Column	Data Type	Missing %	Unique	Role	Treatment
asset_id	int64	0.0	50000	Identifier	Exclude from model to avoid memorization/high-cardinality noise.
asset_type	object	0.0	7	Predictor	Retain; preprocessing based on data type.
asset_age_years	int64	0.0	39	Predictor	Retain; preprocessing based on data type.
substation_region	object	0.0	5	Predictor	Retain; preprocessing based on data type.
manufacturer	object	0.0	5	Predictor	Retain; preprocessing based on data type.
power_load_mw	float64	0.0	50000	Predictor	Retain; preprocessing based on data type.
load_utilization_pct	float64	0.0	50000	Predictor	Retain; preprocessing based on data type.
voltage_fluctuation_pct	float64	0.0	50000	Predictor	Retain; preprocessing based on data type.
frequency_deviation_hz	float64	0.0	50000	Predictor	Retain; preprocessing based on data type.
transformer	float64	0.0	49985	Predictor	Retain;

_temperature_c					preprocessing based on data type.
oil_quality_score	float64	9.996	45010	Predictor	Retain; preprocessing based on data type. Missing values: median imputation with missingness indicator.
equipment_health_score	float64	0.0	50000	Predictor	Retain; preprocessing based on data type.
vibration_level	float64	0.0	50000	Predictor	Retain; preprocessing based on data type.
maintenance_overdue_days	int64	0.0	180	Predictor	Retain; preprocessing based on data type.
maintenance_events_last_12m	int64	0.0	14	Predictor	Retain; preprocessing based on data type.
inspection_score	float64	10.117	44953	Predictor	Retain; preprocessing based on data type. Missing values: median imputation with missingness indicator.
previous_outages_12m	int64	0.0	11	Predictor	Retain; preprocessing based on data type.
avg_outage_duration_mi	float64	0.0	50000	Leakage / impact	Exclude from model;

nutes					retain for impact analysis only.
customer_complaints_last_12m	int64	0.0	17	Predictor	Retain; preprocessing based on data type.
temperature_c	float64	0.0	50000	Predictor	Retain; preprocessing based on data type.
humidity_pct	float64	0.0	50000	Predictor	Retain; preprocessing based on data type.
wind_speed_kmh	float64	0.0	50000	Predictor	Retain; preprocessing based on data type.
rainfall_mm	float64	0.0	50000	Predictor	Retain; preprocessing based on data type.
storm_risk_index	float64	0.0	50000	Predictor	Retain; preprocessing based on data type.
customers_served	int64	0.0	39431	Business impact	Exclude from predictive model; use for maintenance priority.
estimated_revenue_loss	float64	0.0	49901	Leakage / impact	Exclude from model; retain for impact analysis only.
regulatory_penalty_cost	float64	0.0	50000	Leakage / impact	Exclude from model; retain for impact analysis only.

legacy_asset_code	int64	0.0	48635	Identifier	Exclude from model to avoid memorization/high-cardinality noise.
grid_cluster_id	int64	0.0	49	Predictor	Retain; preprocessing based on data type.
monitoring_batch_id	int64	0.0	8964	Identifier	Exclude from model to avoid memorization/high-cardinality noise.
administrative_reference	int64	0.0	48594	Identifier	Exclude from model to avoid memorization/high-cardinality noise.
grid_failure_flag	int64	0.0	2	Target	Binary target; validate 0/1.

4. Data Quality and Leakage

Exact duplicate rows were removed. Categorical text values were normalized by stripping whitespace and title-casing strings; this corrects the three observed representations of Transformer/transformer/TRANSFORMER. Missing numerical values are handled inside the model pipeline using median imputation plus missingness indicators. Categorical variables use most-frequent imputation and one-hot encoding.

Leakage Assessment

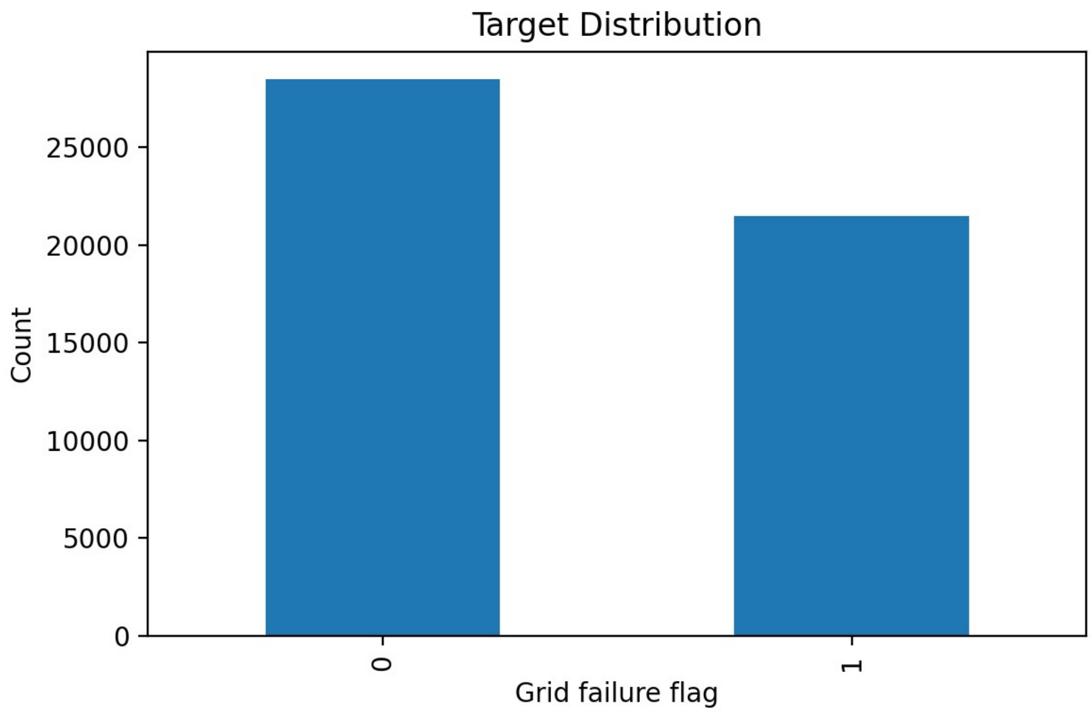
Feature	Risk	Reason	Decision
estimated_revenue_loss	High	May be calculated from outage/failure consequences.	Exclude from predictive features; consider for retrospective business impact if decision-stage availability is

			confirmed.
regulatory_penalty_cost	High	May arise after a reportable outage/failure.	Exclude from predictive features; consider for retrospective business impact if decision-stage availability is confirmed.
avg_outage_duration_minutes	High	Outage duration is an event outcome and may be known only after failure.	Exclude from predictive features; consider for retrospective business impact if decision-stage availability is confirmed.
asset_id	Medium	Identifier/reference field; high cardinality can cause memorization without causal signal.	Exclude from model.
legacy_asset_code	Medium	Identifier/reference field; high cardinality can cause memorization without causal signal.	Exclude from model.
monitoring_batch_id	Medium	Identifier/reference field; high cardinality can cause memorization without causal signal.	Exclude from model.
administrative_reference	Medium	Identifier/reference field; high cardinality can cause memorization without causal signal.	Exclude from model.

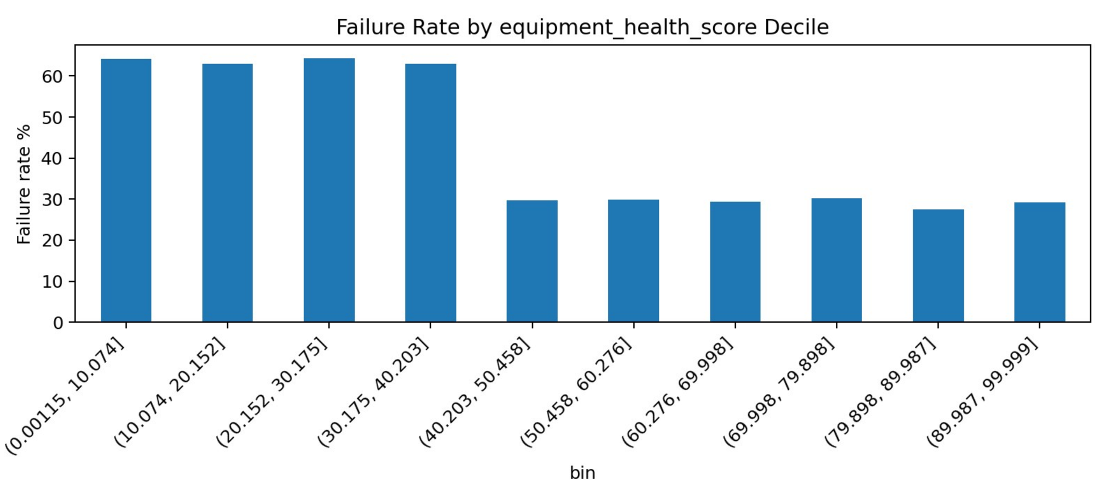
customers_served is excluded from predictive modeling and retained only for business-impact prioritization. Identifier/reference columns are excluded to avoid memorization and spurious high-cardinality effects.

5. Exploratory Data Analysis

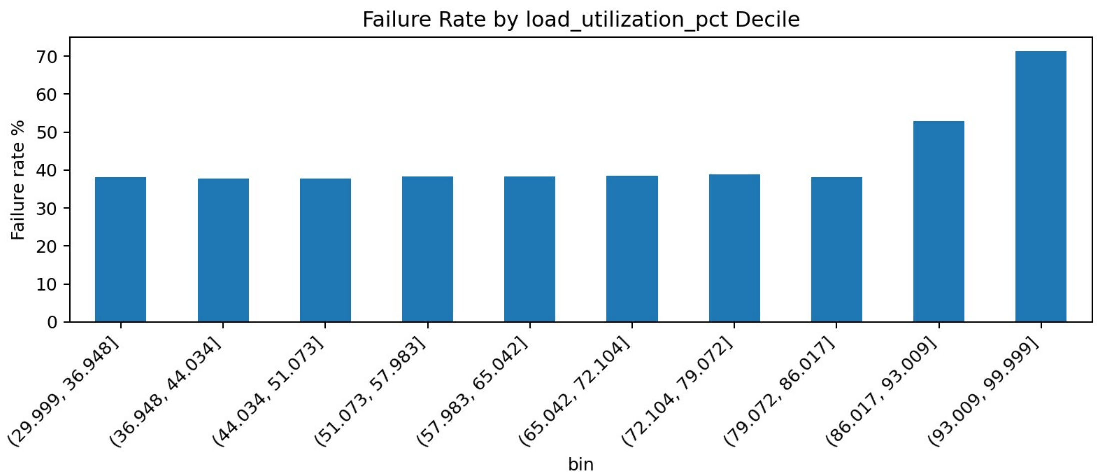
Failure behavior is especially differentiated by equipment condition, weather stress, electrical stability, and loading. The strongest univariate patterns include much higher failure rates at low equipment-health scores, very high load utilization, and high storm-risk levels. Asset-type failure rates are relatively similar after normalizing inconsistent Transformer capitalization, so asset type alone is not a dominant discriminator.



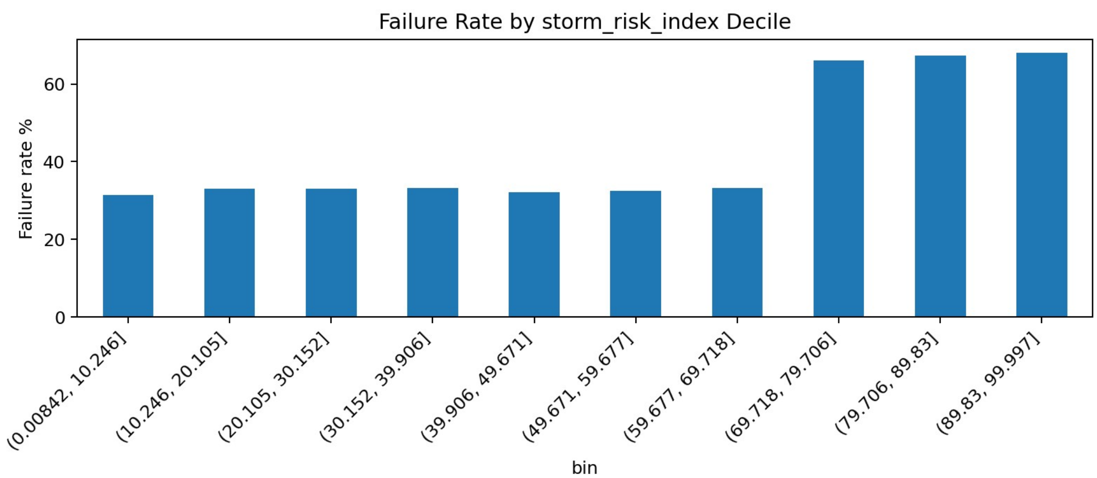
Target distribution



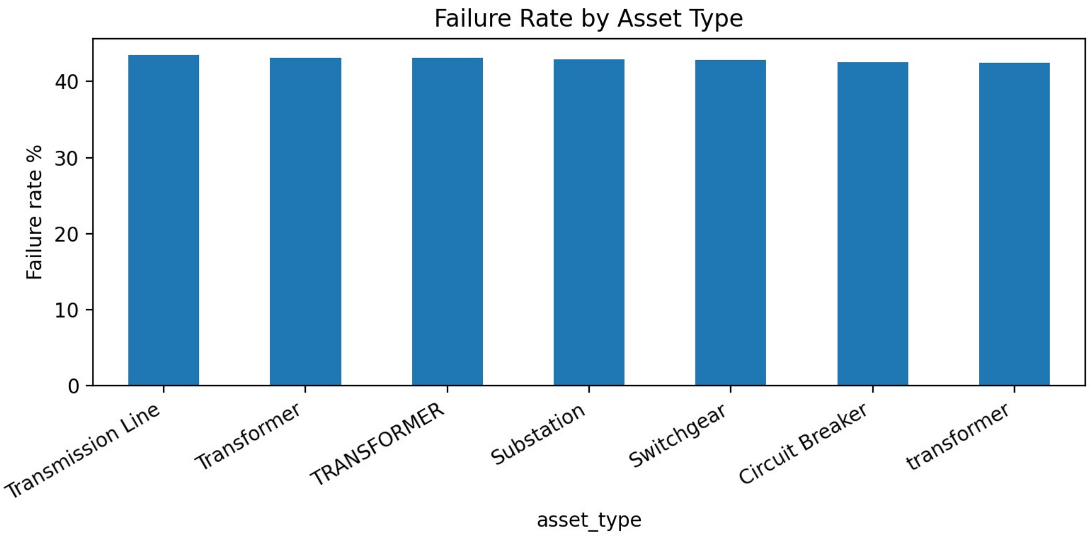
Failure rate across equipment-health deciles



Failure rate across load-utilization deciles



Failure rate across storm-risk deciles



Failure rate by normalized asset type

6. Feature Engineering

high_load_flag: Flags load utilization >=85% to identify operational stress.

poor_health_flag: Flags equipment health <40 to identify degraded condition.

maintenance_overdue_flag: Flags maintenance overdue >90 days.

high_storm_risk_flag: Flags storm risk >=75.

load_stress_index: Combines power load and utilization into a simple stress measure.

age_health_risk: Combines asset age with declining health into a condition-aging indicator.

7. Model Development and Evaluation

The dataset is split stratified into training, validation, and test partitions. Class weights are used rather than SMOTE because the target is moderately imbalanced rather than extremely rare and the objective is to preserve the observed data distribution. The validation set is used to choose an operating threshold that emphasizes recall through F2 optimization.

Model	Precision	Recall	F1	ROC-AUC	PR-AUC
Random Forest	0.660	0.943	0.777	0.911	0.886
Decision Tree	0.637	0.944	0.761	0.899	0.862
SVM	0.565	0.960	0.712	0.871	0.841

Logistic Regression	0.581	0.947	0.720	0.871	0.841

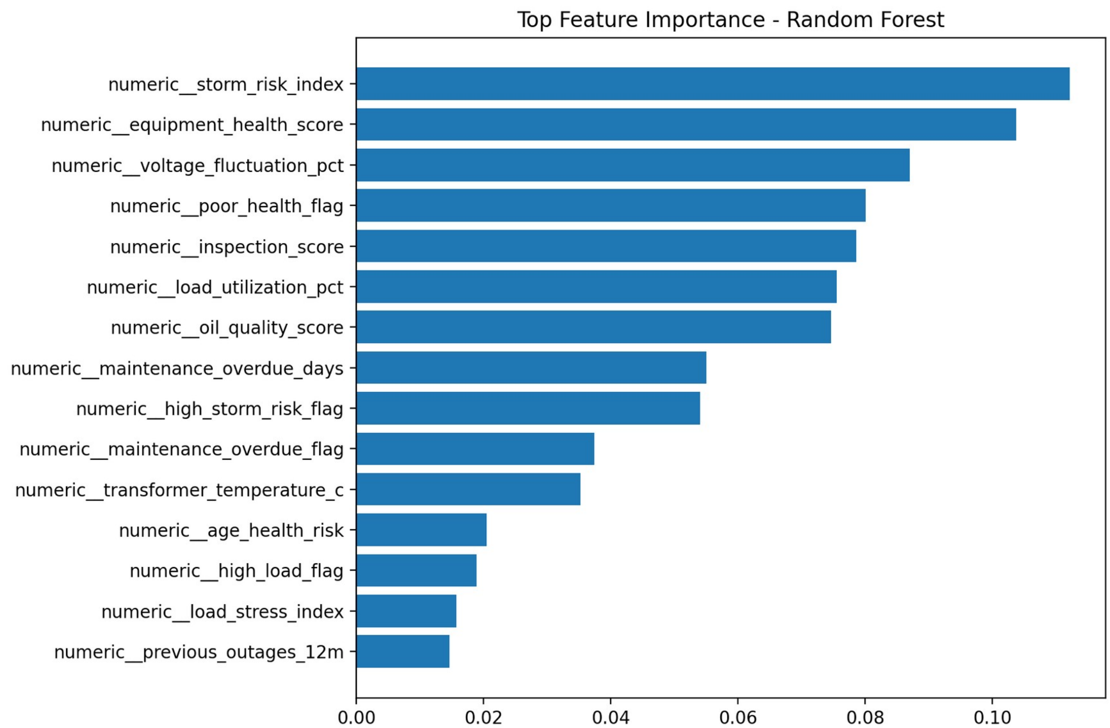
Selected model: Random Forest. Test PR-AUC is the primary selection criterion because the project is a failure-risk classification problem where ranking positive cases is operationally important. Recall remains a key business metric because false negatives can lead to outages and emergency intervention.

SVM note: the SVM implementation uses LinearSVC and a sigmoid transformation of its decision score for ranking/metric calculations. These values should not be interpreted as calibrated probabilities. The final selected Random Forest produces genuine class probabilities through predict_proba().

8. Important Failure Drivers

- storm_risk_index: importance 0.1122
- equipment_health_score: importance 0.1037
- voltage_fluctuation_pct: importance 0.0870
- poor_health_flag: importance 0.0801
- inspection_score: importance 0.0786
- load_utilization_pct: importance 0.0755
- oil_quality_score: importance 0.0747
- maintenance_overdue_days: importance 0.0551
- high_storm_risk_flag: importance 0.0540
- maintenance_overdue_flag: importance 0.0374

The leading model features are storm risk, equipment health, voltage fluctuation, inspection score, load utilization, oil quality, and maintenance overdue days. These should be interpreted as predictive associations, not causal effects.



9. Risk Scoring and Maintenance Priority

The Random Forest outputs failure_probability. Business-facing categories are Low (<30%), Medium (30–60%), High (60–80%), and Critical (>=80%). Separately, a validation-selected classification threshold of 0.29 is used for binary intervention decisions because it provides a strong recall/precision trade-off. Maintenance priority combines predicted risk with normalized customers_served; customer exposure is not used as a predictor.

Top 10 Prioritized Assets

Rank	Asset ID	Type	Region	Failure Prob.	Risk	Customers	Priority	Recommended Action
1	139691	Substation	Central	0.997	Critical	99675	99.52	Immediate engineering review
2	113625	Transmission Line	Central	0.998	Critical	99443	99.50	Immediate engineering review
3	101155	Transfo	Central	0.994	Critical	99774	99.30	Immed

		rmer						iate engineering review
4	119357	Circuit Breaker	West	0.996	Critical	99444	99.28	Immediate engineering review
5	117865	Transformer	South	0.995	Critical	99002	99.00	Immediate engineering review
6	134864	Substation	North	0.994	Critical	98979	98.85	Immediate engineering review
7	109138	Circuit Breaker	East	0.990	Critical	99500	98.77	Immediate engineering review
8	142176	Switchgear	West	0.991	Critical	99117	98.71	Immediate engineering review
9	144149	Switchgear	Central	0.991	Critical	98889	98.57	Immediate engineering review
10	139413	Transmission Line	South	0.993	Critical	98285	98.47	Immediate engineering review

10. Business Recommendations

- Inspect Critical-risk assets first, especially assets combining high model probability with large customer exposure.
- Use equipment-health, inspection, oil-quality, voltage-fluctuation, and maintenance-overdue indicators as condition-monitoring triggers.

- Increase operational monitoring during high storm-risk periods and review asset loading when utilization approaches or exceeds 85%.
- Use the ranked asset list in weekly maintenance planning, with engineering review for Critical assets and scheduled intervention for High assets.
- Integrate the score into an asset-management or CMMS workflow and record inspection outcomes so future model retraining can be evaluated against real interventions.

11. Limitations

- No reliable timestamp means temporal validation and explicit prediction horizons cannot be established.
- The dataset does not establish causality; model importance indicates association.
- Post-event/business-consequence fields were excluded from predictive modeling because their decision-stage availability is uncertain.
- The supplied dataset should be validated against real utility data, operational processes, and maintenance outcomes before production use.
- SVM scores are not calibrated probabilities in this implementation.

12. Future Scope

- Time-series and survival analysis
- Real-time IoT/smart-grid telemetry
- Weather forecast integration
- Automated maintenance alerts
- MLOps monitoring and drift detection
- Integration with utility asset-management systems

13. Reproducibility

Install dependencies with pip install -r requirements.txt. Run python src/audit.py, python src/eda.py, and python src/train.py. To score new data, run python src/test_new_dataset.py --input <path-to-new-file>. The trained Random Forest, preprocessing pipeline, threshold configuration, metrics, figures, and scored asset tables are saved under models/ and outputs/.